

Class_19_lab

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Pertussis is a bacterial lung infection originally treated with a wP attenuated vaccine. This lab will start with importing case number data from the cdc.

```
cdc <- data.frame(
  Year = c(1922L,
    1923L, 1924L, 1925L, 1926L, 1927L, 1928L,
    1929L, 1930L, 1931L, 1932L, 1933L, 1934L, 1935L,
    1936L, 1937L, 1938L, 1939L, 1940L, 1941L,
    1942L, 1943L, 1944L, 1945L, 1946L, 1947L, 1948L,
    1949L, 1950L, 1951L, 1952L, 1953L, 1954L,
    1955L, 1956L, 1957L, 1958L, 1959L, 1960L,
    1961L, 1962L, 1963L, 1964L, 1965L, 1966L, 1967L,
    1968L, 1969L, 1970L, 1971L, 1972L, 1973L,
    1974L, 1975L, 1976L, 1977L, 1978L, 1979L, 1980L,
    1981L, 1982L, 1983L, 1984L, 1985L, 1986L,
    1987L, 1988L, 1989L, 1990L, 1991L, 1992L, 1993L,
    1994L, 1995L, 1996L, 1997L, 1998L, 1999L,
    2000L, 2001L, 2002L, 2003L, 2004L, 2005L,
    2006L, 2007L, 2008L, 2009L, 2010L, 2011L, 2012L,
    2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
    2019L, 2020L, 2021L, 2022L, 2023L, 2024L),
  Cases = c(107473,
    164191, 165418, 152003, 202210, 181411,
    161799, 197371, 166914, 172559, 215343, 179135,
    265269, 180518, 147237, 214652, 227319, 103188,
    183866, 222202, 191383, 191890, 109873,
    133792, 109860, 156517, 74715, 69479, 120718,
    68687, 45030, 37129, 60886, 62786, 31732, 28295,
    32148, 40005, 14809, 11468, 17749, 17135,
    13005, 6799, 7717, 9718, 4810, 3285, 4249,
    3036, 3287, 1759, 2402, 1738, 1010, 2177, 2063,
```

```

1623,1730,1248,1895,2463,2276,3589,
4195,2823,3450,4157,4570,2719,4083,6586,
4617,5137,7796,6564,7405,7298,7867,
7580,9771,11647,25827,25616,15632,10454,
13278,16858,27550,18719,48277,28639,
32971,20762,17972,18975,15609,18617,6124,
2116,3044,7063,22538)
)

library(ggplot2)

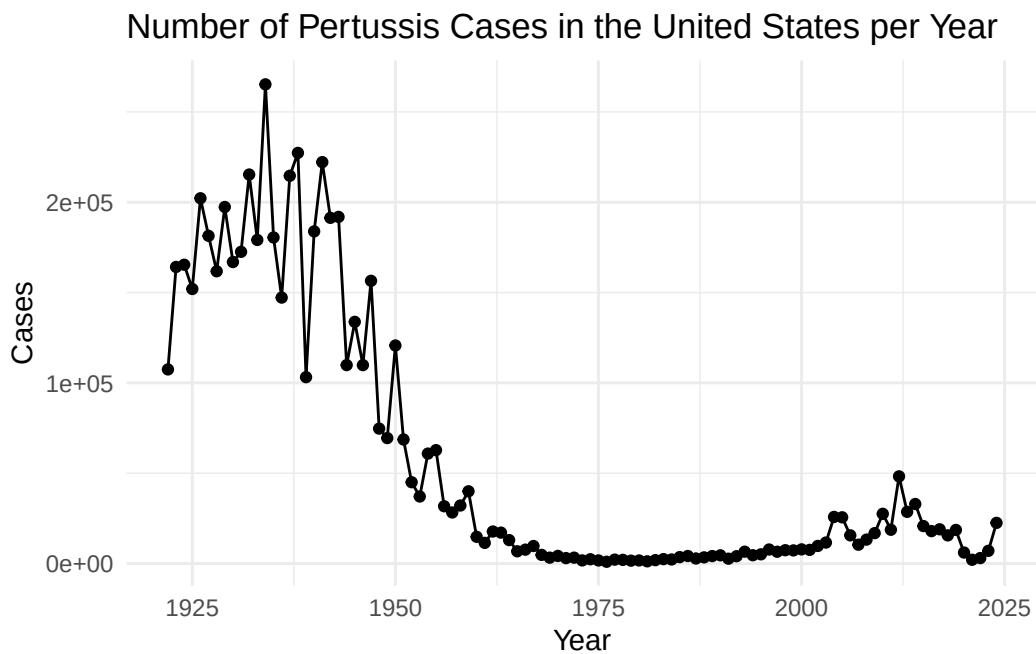
```

Warning: package 'ggplot2' was built under R version 4.5.2

```

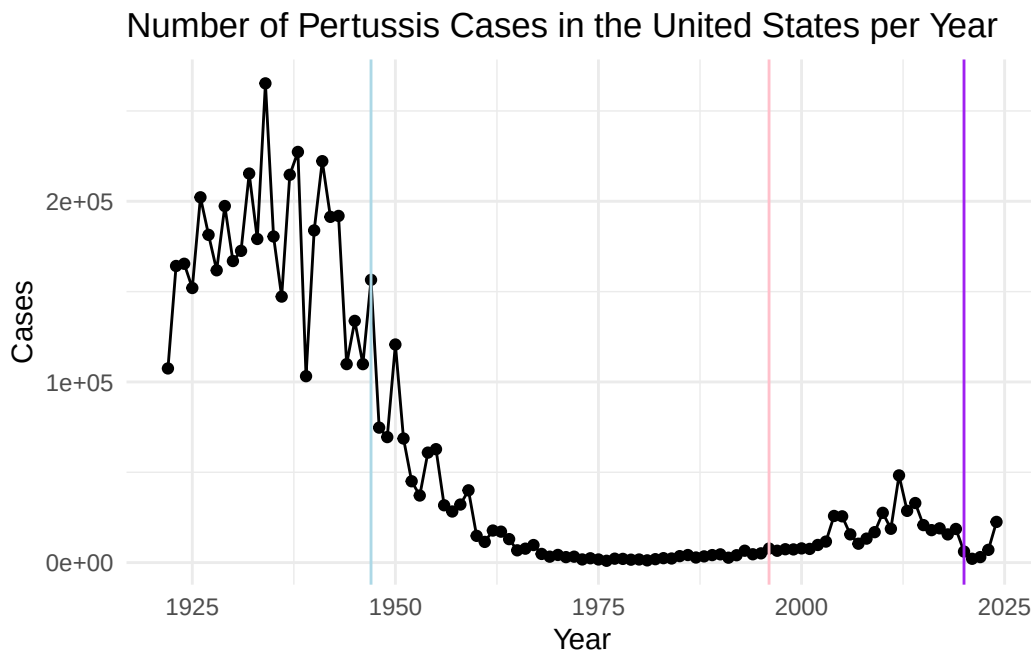
ggplot(cdc) +
  aes(Year, Cases) +
  geom_point() +
  geom_line() +
  labs(title = "Number of Pertussis Cases in the United States per Year") +
  theme_minimal()

```



Adding major milestones

```
ggplot(cdc) +
  aes(Year, Cases) +
  geom_point() +
  geom_line() +
  labs(title = "Number of Pertussis Cases in the United States per Year") +
  theme_minimal() +
  geom_vline(xintercept = 1947, col = "lightblue") +
  geom_vline(xintercept = 1996, col = "pink") +
  geom_vline(xintercept = 2020, col = "purple")
```



The full introduction of the wP (whole cell attenuated) vaccine caused a dramatic decrease in pertussis cases starting in 1947. 10 years after the introduction of the aP (acellular vaccine) there has been an increase in the number of cases annually. This could be due to anti-vax movements causing distrust in public health and a lower vaccination rate in the population. It could also just be that the a-cellular vaccine is lacking in certain components that generated a strong immune response in the wP vaccine. The slow rise could be because individuals with the older vaccine were still protected. Something about the new aP vaccine does not confer lasting immunity, thus needing boosters and why we see a 10 year pause before cases increase. During social distancing due to covid-19 pandemic in 2020 cases dropped but in recent years the number of cases has been rising to pre-pandemic levels.

CMI-PB project

The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)

library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Q. How many wP and aP subjects are there?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q. What is the breakdown by “biological_sex” and “race”?

```
table(subject$biological_sex)
```

```
Female  Male
  112    60
```

```
table(subject$race)
```

```

American Indian/Alaska Native
      1
Asian
     44
Black or African American
      5
More Than One Race
     19
Native Hawaiian or Other Pacific Islander
      2
Unknown or Not Reported
     21
White
     80
```

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown of demographics is not a great representation of the US population as a whole, but it's what we have to work with.

```
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = T)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = T)
```

Now that we have the data imported we can combine our datasets to analyze the whole data.

```
meta <- inner_join(subject, specimen)
```

Joining with `by = join\_by(subject\_id)`

```
abData <- inner_join(ab_titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q. How many Ab measurements do we have?

```
nrow(abData)
```

```
[1] 61956
```

Q How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abData$isotype)
```

```
 IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  7265 11993 12000 12000 12000
```

Q. how many different antigens?

```
16
```

```
unique(abData$antigen)
```

```
[1] "Total"  "PT"      "PRN"      "FHA"      "ACT"      "LOS"      "FELD1"
[8] "BETV1"  "LOLP1"   "Measles"  "PTM"      "FIM2/3"   "TT"       "DT"
[15] "OVA"    "PD1"
```

Q. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abData$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520         8085         7301         15050
```

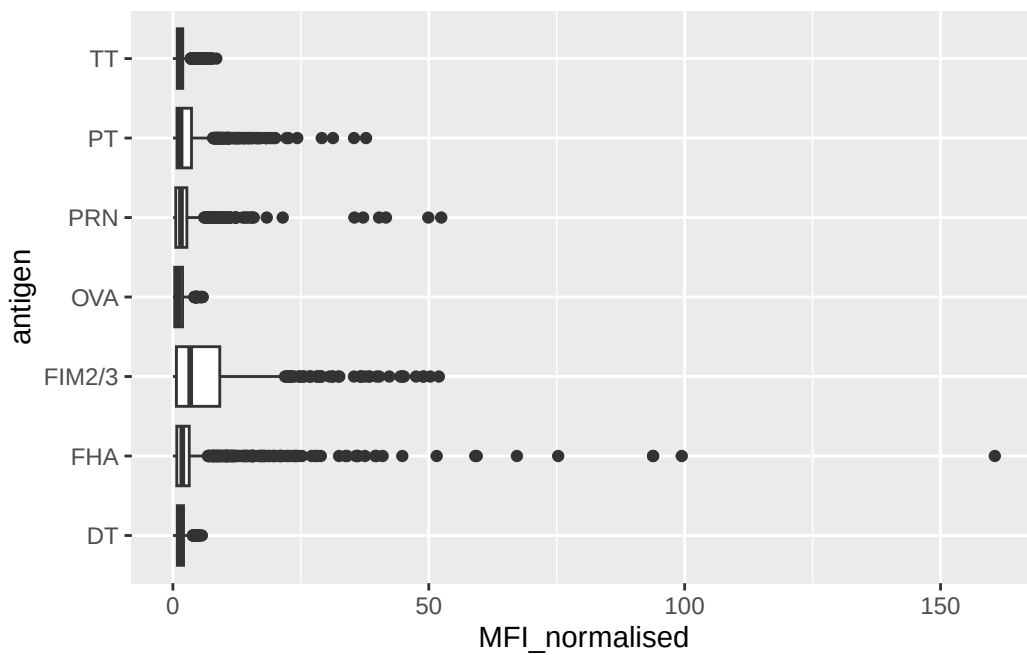
The most current dataset has about half as many data points as the initial 2020 dataset but double that of the 2021 or 2022 datasets.

Examining IgG Ab titer levels

IgG is the most crucial antibody for lasting immunity.

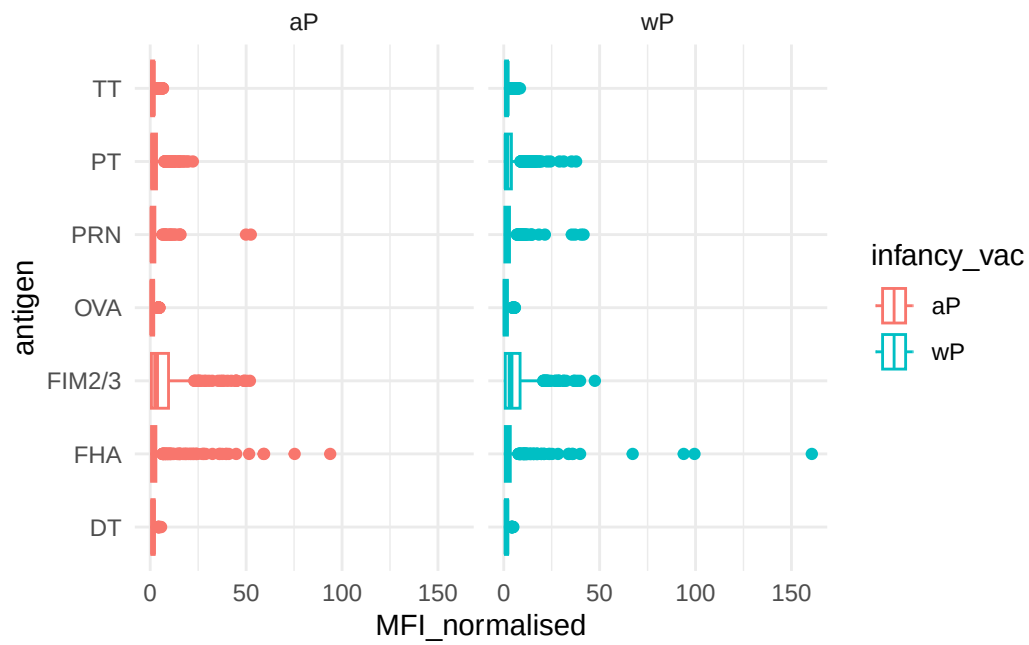
```
IgG <- abData |> filter(isotype == "IgG")

ggplot(IgG) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



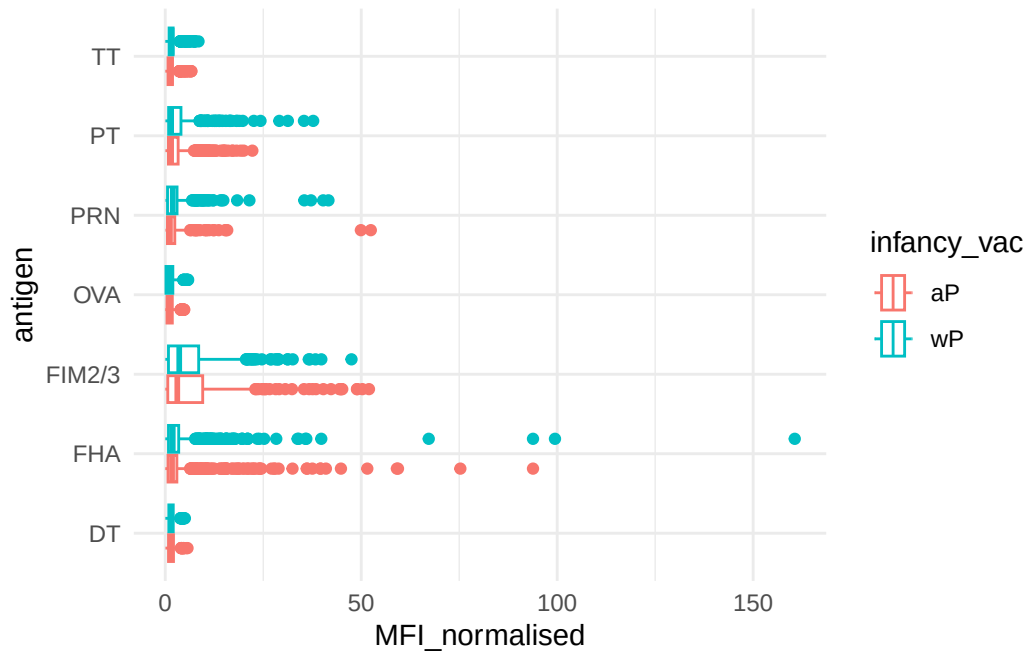
Faceting the data by wP vs aP

```
ggplot(IgG) +
  aes(MFI_normalised, antigen, col= infancy_vac) +
  geom_boxplot() +
  facet_wrap(~infancy_vac) +
  theme_minimal()
```



or

```
ggplot(IgG) +
  aes(MFI_normalised, antigen, col= infancy_vac) +
  geom_boxplot() +
  theme_minimal()
```

focusing on the main pT toxin

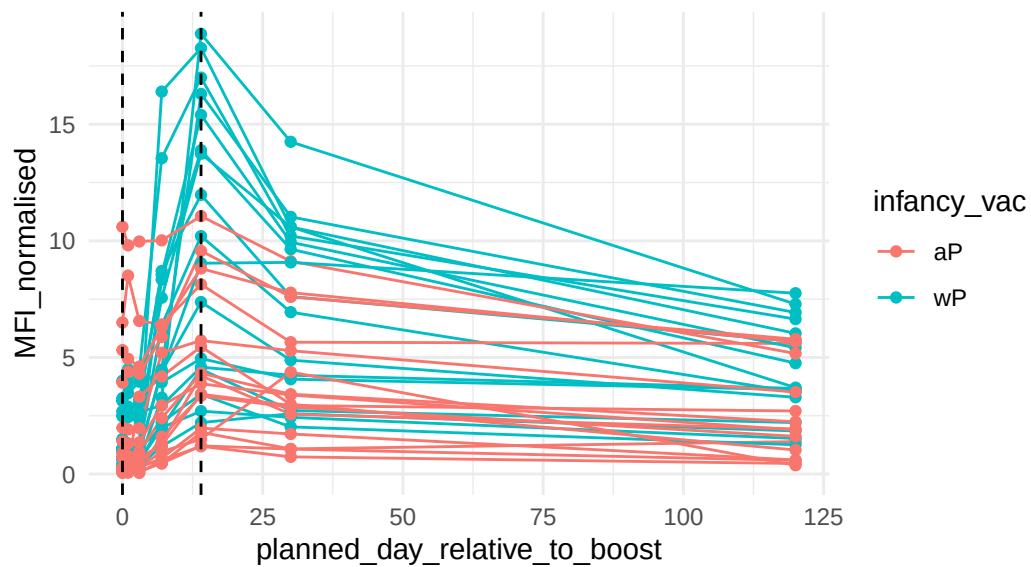
```
pT <- IgG |> filter(antigen == "PT")

pT.21 <- pT |> filter(dataset == "2021_dataset")

ggplot(pT.21) +
  aes(planned_day_relative_to_boost, MFI_normalised,
      col = infancy_vac, group = subject_id) +
  geom_line() +
  geom_point() +
  theme_minimal() +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 14, linetype = "dashed") +
  labs(title = "2021 dataset IgG antibody pertussis Toxin Antigen", subtitle = "dashed lines")
```

2021 dataset IgG antibody pertussis Toxin Antigen

dashed lines represent exposure event and peak of antibody response



This graph shows the time course of antibody response to pertussis toxin comparing the wP vaccine (teal) to the aP vaccine (red). The levels peak around day 14 and decline and in general the whole bacteria vaccine (wP) has a higher level of antibody response.